

A Packing of 5 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 5 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 3.181627. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.054382, obtained by deleting 1 of the 6 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 5 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.181627 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 5$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.054382, obtained by deleting 1 of the 6 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 5/4 = 3.181818$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N5.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
318162796472152591134765800277340641268692829003268991859654019329966879
151974027337368448908412416832869588795914614481873837517053194448845806
7233659050049313021988057516116271250657787373255
```

denominator:

```
999999999998381501371550181099330017048005865072022225809003764278098908
486129747081590891343841654115048018580402964714867686628954495901661724
092171457035276449706081162230012939538931267824
```

B Coordinates

5 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N5.txt`.

```
405203867040 368715820940 225812786649 366078748914 18816241838 312877762732 128632156541 210541295289 99614026691 213120582606 -12154623547 494068844800
28971455906 -54506060168 336341192427 -387781875145 13435300075 -141903507168 364519445423 26426860077 2929279095874 -700279624726 166187697940 237662076315
77788590053 84448850970 -207968601052 -504001934835 33535114950 -212347787911 -134881427379 144461586342 2611844588 -149850730395 407833570501 92427066027
34858635096 -267629801441 -277068191175 -27664194918 12254770003 324072693179 -149536510575 220340502960 34765776940 -49843045555 476990962884 15547026778
-640293465111 -369897961525 150264241870 190956592995 3891339489 -197330594044 -144614958866 -36335738485 7064420901 -268494407556 -183138301267 374517567249
-208020338154 121082105704 524615991837 -44706126471 52828314164 242402523395 543968584911 23268601444 15891565346 311974204286 85861712305 193383829110
-190193062613 65059277124 -145808463377 -294855613359 4765836907 203900094529 -287205968788 -69266121325 180722774 -4271805313 -602055096125 -176916155891
-106102029034 -53514959689 541044766118 -2075772497 1021898253 507350636686 121968684006 96175158627 277557725035 311143985949 31507365549 -176038929459
282923614606 11362424058 224114771175 -442255214371 128103140058 -69606725081 368954604474 -51272023593 15755971013 -346701739743 -107500028934 10110715370
-99075422980 478872754530 86804926693 -16938867846 456042034087 169175299023 -558081823624 -130339767 136192595235 209347814942 100490327408 -161519168961
-486994205688 627925358080 -215202490481 -143857718435 674655361 -444913186437 -103494492659 652512171 1522346506381 458391414697 -219931492207 196551751876
211154207892 -87302985886 538476935578 -24823057719 53051855847 15690950057 124536211613 -168479832323 74369270064 -640683898469 -177546743862 24763464188
-148985725641 104796219734 6292134758 679207003442 139794692488 380906621160 132472733453 373146366062 174063668118 -102799648351 -160430440946 251758344810
-342123779329 27152500414 410227606125 28747961915 156716507886 -348639418809 268528714995 93130261996 218921566744 45162220080 144361695828 -432362021214
236969175155 -666518780065 306579781196 -19209538535 1651696018 56857410163 151227645524 -389019955496 204280145459 96167494993 223808663284 755611660708
140591373210 502115373514 387345637591 -1040304279 6298547603511 -176108341483 -615729424949 -74415271 1231467272412 -127699496830 641679652830 -190348036369
-267654623495 44667493490 89563927917 -263080590357 59503869994 271559280937 102216438807 -29311106514 11255248482 283515922819 -162998028454 -121548105505
-305195312311 -54306756851 327278571102 -111840569 732007185391 338440247375 154479208694 485911289055
66319144954 340920017899 122237012962 -436966558485 28763245505 217481460608 -318070576208 184579230335
```